

2.Common commands

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2.1.Do not use sudo command

Normally, to operate docker commands, you need to prefix sudo, as follows:

```
sudo docker version
```

But after adding the docker user group, you don't need to add the sudo prefix. How to add the docker user group (run the command in the host running docker):

```
sudo groupadd docker          # Add docker user group
sudo gpasswd -a $USER docker  # Add the current user to the docker user group,
where $USER can automatically resolve to the currently logged in user
newgrp docker                 # Update docker user group
```

After adding the above command, use the [docker images] command to test. If no error is reported, it means that the sudo command is no longer needed. If the following error is reported:

```
pi@ubuntu:~$ docker images
WARNING: Error loading config file: /home/pi/.docker/config.json: open
/home/pi/.docker/config.json: permission denied
```

Then execute the following command on the host machine to solve the problem:

```
sudo chown "$USER":"$USER" /home/"$USER"/.docker -R
sudo chmod g+rwX "/home/$USER/.docker" -R
```

2.2. Help command

```
docker info    #Display Docker system information, including the number of images
and containers.
docker --help  #Help
```

2.3. Image command

1. docker pull download image

```
# Download image
jetson@ubuntu:~$ docker pull ubuntu
Using default tag: latest                # If no tag is specified, the
default is latest.
latest: Pulling from library/ubuntu
cd741b12a7ea: Pull complete             #Layered download
Digest: sha256:67211c14fa74f070d27cc59d69a7fa9aeff8e28ea118ef3babcb295a0428a6d21
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest         #Real location
```

2. docker images list images

```
# List images on localhost
pi@raspberrypi:~ $ docker images
```

			i	Info	
→	U	In Use			
IMAGE			ID	DISK USAGE	CONTENT
SIZE	EXTRA				
192.168.2.51:5000/muto-humble:3.3.0			23a4c47d106f	7.92GB	
OB					
192.168.2.51:5000/muto-humble:latest			82136c6d9c27	8.34GB	
OB	U				
192.168.2.51:5000/muto-kilted:1.1			2753e1b11ddd	7.4GB	
OB					
192.168.2.51:5000/ros2-kilted-release:1.0			e6252585561e	5.7GB	
OB					
alpine/curl:latest			af0461b4a8c6	14.3MB	
OB					
busybox:latest			eade5be814e8	4.17MB	
OB	U				
kilted:1.0			80471fea8516	7.22GB	
OB					
kilted:1.1			2753e1b11ddd	7.4GB	
OB					
langgenius/dify-api:1.11.1			c5c633a63f56	2.39GB	
OB	U				
langgenius/dify-plugin-daemon:0.5.1-local			3cd80a202e1b	1.43GB	
OB	U				
langgenius/dify-sandbox:0.2.12			5c2ad4db0b19	636MB	
OB	U				
langgenius/dify-web:1.11.1			baad4c046f12	392MB	
OB	U				
muto-humble:3.2.0			7a8ded22e9e9	7.92GB	
OB					

```
# explanation
REPOSITORY Image's warehouse source
TAG image tags
IMAGE ID image ID
CREATED image creation time
```

SIZE Image Size

#The same warehouse source can have multiple TAGs, representing different versions of the warehouse source. We use REPOSITORY: TAG to define different images. If you do not define the tag version of the image, docker will use the latest image by default!

#Optional

-a: List all local images
-q: Show only image id
--digests: Display image summary information

3. docker search search image

```
# Search for images
jetson@ubuntu:~$ docker search ros2
```

NAME	STARS	OFFICIAL	AUTOMATED	DESCRIPTION
osrf/ros2				**Experimental** Docker Images
for ROS2 deve...	60		[OK]	
tiryoh/ros2-desktop-vnc				A Docker image to provide HTML5
VNC interfac...	11			
althack/ros2				An assortment of development
containers for ...	7			
tiryoh/ros2				unofficial ROS2 image
	6			
athackst/ros2				[Deprecated-> use althack/ros2]
	5			
uobflightlabstarling/starling-mavros2				ROS2 version of MAVROS
	2			
theosakamg7/ros2_java_docker				Image base
	1		[OK]	

```
# docker search The name of a certain image corresponds to the image in the
DockerHub warehouse
# Optional
--filter=stars=50 : List the images whose collection number is not less than the
specified value.
```

4. docker rmi delete image

```
# delete image
docker rmi -f image id # delete single
docker rmi -f Image name:tag Image name:tag # Delete multiple
docker rmi -f $(docker images -qa) # Delete all
```

2.4. Container command

A container can only be created if you have an image. We use the ubuntu image here to test.
Download the image:

1. docker run runs the image to start the container

```
# command
docker run [OPTIONS] IMAGE [COMMAND][ARG...]
# Description of common parameters
--name="Name" # Assign a name to the container
-d # Run the container in the background and return the container's id!
-i # Run the container in interactive mode by using it with -t
-t # Reassign a terminal to the container, usually used with -i
-P # Random port mapping (uppercase)
-p # Specify port mapping (summary), generally there are four ways to write it
ip:hostPort:containerPort
ip::containerPort
hostPort:containerPort (commonly used)
containerPort

#test
jetson@ubuntu:~$ docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
yahboomtechnology/ros-humble  3.4.0       49581aa78b6b  2 hours ago   24.3GB
yahboomtechnology/ros-humble  3.3.9       cefb5ac2ca02  3 days ago    20.5GB
yahboomtechnology/ros-humble  3.3.8       49996806c64a  4 days ago    20.5GB
yahboomtechnology/ros-humble  3.3.7       8989b8860d17  4 days ago    17.1GB
yahboomtechnology/ros-humble  3.3.6       326531363d6e  5 days ago    16.1GB
ubuntu               latest       bab8ce5c00ca  6 weeks ago   69.2MB
hello-world          latest       46331d942d63  13 months ago 9.14kB

#Use ubuntu to start the container in interactive mode and execute the /bin/bash
command in the container!
jetson@ubuntu:~$ docker run -it ubuntu:latest /bin/bash
root@c54bf9efae47:/# ls
bin boot dev etc home lib media mnt opt proc root run sbin srv sys
tmp usr var
root@c54bf9efae47:/# exit      # Use exit to exit the container and return to
the host
exit
jetson@ubuntu:~$
```

2. docker ps lists all running containers

```
#command
docker ps [OPTIONS]

#Description of common parameters
-a #List all currently running containers + historically run containers
-l # Show recently created containers
-n=? # Display the last n created containers
-q # Silent mode, only the container number is displayed.

#test
jetson@ubuntu:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
	PORTS NAMES			
c54bf9efae47	ubuntu:latest	"/bin/bash"	2 hours ago	Up 4 seconds
	funny_hugle			
3b9c01839579	hello-world	"/hello"	3 hours ago	Exited (0) 3 hours ago
	jovial_brown			

3. Exit the container

```
exit          # Container stops exiting
ctrl+P+Q     # Container does not stop exiting
```

4. Enter the running container from multiple terminals

```
# command 1
docker exec -it container id bashShell

# test
jetson@ubuntu:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
	PORTS NAMES			
c54bf9efae47	ubuntu:latest	"/bin/bash"	2 hours ago	Up 4 seconds
	funny_hugle			
3b9c01839579	hello-world	"/hello"	3 hours ago	Exited (0) 3 hours ago
	jovial_brown			

```
jetson@ubuntu:~$ docker exec -it c5 /bin/bash # The ID of the container can be
abbreviated, as long as it can uniquely identify the container.
root@c54bf9efae47:/#

# command 2
docker attach container id

# test
jetson@ubuntu:~$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
	PORTS NAMES			
c54bf9efae47	ubuntu:latest	"/bin/bash"	2 hours ago	Up 35 seconds
	funny_hugle			
3b9c01839579	hello-world	"/hello"	3 hours ago	Exited (0) 3 hours ago
	jovial_brown			

```
jetson@ubuntu:~$ docker attach c5 # The ID of the container can be
abbreviated, as long as it can uniquely identify the container.
```

```
root@c54bf9efae47:/#
```

```
#Distinction
```

```
# exec opens a new terminal in the container and can start a new process
```

```
# attach directly enters the terminal of the container startup command and will not start a new process
```

5. Start and stop the container

```
docker start (container id or Container name)    # Start container
docker restart (container id or Container name)  # Restart container
docker stop (container id or Container name)     # Stop container
docker kill (container id or Container name)     # Force stop container
```

6. Delete containers

```
docker rm container id                # delete specified container
docker rm -f $(docker ps -a -q)       # Delete all containers
docker ps -a -q|xargs docker rm       # Delete all containers
```

2.5. Other commonly used commands

1. View the process information running in the container and support ps command parameters.

```
# command
docker top container id

# test
jetson@ubuntu:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS         NAMES
c54bf9efae47   ubuntu:latest  "/bin/bash"            2 hours ago   Up 2 minutes
              funny_hugle
3b9c01839579   hello-world    "/hello"                3 hours ago   Exited (0) 3 hours ago
              jovial_brown
jetson@ubuntu:~$ docker top c5
UID            PID            PPID            C
STIME          TTY            TIME            CMD
root           9667           9647            0
14:20          pts/0          00:00:00        /bin/bash
```

2. View the metadata of the container/image

```
# command
docker inspect container id
```

#Test to view container metadata

jetson@ubuntu:~\$ docker ps -a

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
c54bf9efae47	ubuntu:latest	"/bin/bash"	2 hours ago	Up 4 minutes
3b9c01839579	hello-world	"/hello"	3 hours ago	Exited (0) 3 hours ago

jetson@ubuntu:~\$ docker inspect c54bf9efae47

```
[
  {
    # The complete id, the container id above is the first few digits of the
    intercepted id.
    "Id":
    "c54bf9efae471071391202a8718b346d9af76cb1ff17741e206280603d6f0056",
    "Created": "2023-04-24T04:19:46.232822024Z",
    "Path": "/bin/bash",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 9667,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2023-04-24T06:20:58.508213216Z",
      "FinishedAt": "2023-04-24T06:19:45.096483592Z"
    },
  },
]
```

Test to view image metadata

jetson@ubuntu:~\$ docker images

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
ubuntu	latest	bab8ce5c00ca	6 weeks ago	69.2MB
hello-world	latest	46331d942d63	13 months ago	9.14kB

jetson@ubuntu:~\$ docker inspect bab8ce5c00ca

```
[
  {
    "Id":
    "sha256:bab8ce5c00ca3ef91e0d3eb4c6e6d6ec7cffa9574c447fd8d54a8d96e7c1c80e",
    "RepoTags": [
      "ubuntu:latest"
    ],
    "RepoDigests": [
      "ubuntu@sha256:67211c14fa74f070d27cc59d69a7fa9aeff8e28ea118ef3babc295a0428a6d21"
    ],
    "Parent": "",
    "Comment": "",
    "Created": "2023-03-08T04:32:41.063980445Z",
    "Container":
    "094fd0c521be8c84d81524e4a5e814e88a2839899c56f654484d32d171c7195b",
    "ContainerConfig": {

```

```

        "Hostname": "094fd0c521be",
        .....
        "Labels": {
            "org.opencontainers.image.ref.name": "ubuntu",
            "org.opencontainers.image.version": "22.04"
        }
    },
    "DockerVersion": "20.10.12",
    "Author": "",
    "Config": {
        "Hostname": "",
        .....
        "Labels": {
            "org.opencontainers.image.ref.name": "ubuntu",
            "org.opencontainers.image.version": "22.04"
        }
    },
    "Architecture": "arm64",
    "Variant": "v8",
    "Os": "linux",
    "Size": 69212233,
    "VirtualSize": 69212233,
    "GraphDriver": {
        "Data": {
            "MergedDir":
"/var/lib/docker/overlay2/8418b919a02d38a64ab86060969b37b435977e9bbdeb6b0840d4eb
698280e796/merged",
            "UpperDir":
"/var/lib/docker/overlay2/8418b919a02d38a64ab86060969b37b435977e9bbdeb6b0840d4eb
698280e796/diff",
            "WorkDir":
"/var/lib/docker/overlay2/8418b919a02d38a64ab86060969b37b435977e9bbdeb6b0840d4eb
698280e796/work"
        },
        "Name": "overlay2"
    },
    "RootFS": {
        "Type": "layers",
        "Layers": [

"sha256:874b048c963ab55b06939c39d59303fb975d323822a4ea48a02ac8dc635ea371"

        ]
    },
    "Metadata": {
        "LastTagTime": "0001-01-01T00:00:00Z"
    }
}
]

```

2.6. Command summary

